

of Questions: 30

Question #: 1

A 55-year-old woman comes to the physician because of a 2-month history of early satiety, nausea, and vomiting of partially digested food. She also has epigastric fullness, especially after meals, which lessens after vomiting. Imaging studies of the abdomen show small bowel obstruction secondary to an aneurysm of the superior mesenteric artery at its origin from the aorta. Which of the following is most likely the site of compression?

- A. Pyloric region of the stomach
- B. 1st part of the duodenum
- ✓C. 3rd part of the duodenum
- D. Terminal ileum
- E. Ascending colon

Rationale: The superior mesenteric artery crosses over the 3rd/ horizontal part of the duodenum, therefore an aneurysm in this artery will compress on this part.

Question #: 2

A 25-year-old man is brought to the emergency department because of a 2-hour history of severe abdominal pain. Physical examination shows rebound tenderness over the right lower quadrant and a positive Psoas sign. Ultrasonography of the abdomen shows an enlarged appendix with thickened walls. Laparoscopic surgical removal is difficult because there is a fat-filled peritoneal fold, preventing visualization of the appendix. Which of the following is the most likely embryological origin of the structure obscuring the view?

- A. Foregut
- ✓B. Dorsal mesentery
- C. Ventral mesentery
- D. Midgut
- E. Hindgut

Rationale: Fat filled peritoneal fold is greater omentum and is a derivative of dorsal mesentery.

Question #: 3

A 70-year-old woman is brought to the emergency department because of a 4-hour history of intense epigastric abdominal pain that radiates to the back. Physical examination shows jaundice. Endoscopic retrograde pancreatography shows a tumor at the head of the pancreas. Which of the following nervous structures most likely carries the visceral afferents that were stimulated to result in the patient's pain?

- A. Vagus
- B. Lesser splanchnics
- ✓C. Greater splanchnics
- D. Pelvic splanchnics
- E. Least splanchnics

Rationale: The pancreas is a foregut organ, therefore it receives sympathetic supply from T5-T9 via the greater splanchnic nerves. Visceral afferents for pain from the pancreas travel with sympathetics, so they will travel back to the spinal cord with the greater splanchnic nerves.

Question #: 4

A 3-month-old boy is brought to the emergency department by his parents because of a 6-hour history of vomiting. Imaging studies of the abdomen show gastric outlet obstruction due to pyloric stenosis. Which of the following arterial blood gas results is most likely in this patient?

- A. I
- B. II
- C. III
- ✓D. IV
- E. V

Rationale: Pyloric stenosis (gastric outlet syndrome) is characterized by metabolic alkalosis (loss of gastric HCl). Arterial pH is increased; Serum HCO_3^- is increased (loss of H^+ in gastric fluid adds a bicarbonate to the ECF); PCO_2 is elevated as a compensatory response (alkalosis inhibits the respiratory center and results in hypoventilation).

Lab results I and II - may be found in mixed acid-base disorders. Note PCO_2 and HCO_3^- are

moving in opposite directions.

Lab results III - Observed in acute respiratory acidosis. Note the normal HCO_3 levels. In chronic respiratory acidosis, the HCO_3 levels increase.

Lab results V - Observed in acute respiratory alkalosis. Note the normal HCO_3 levels. In chronic respiratory alkalosis, the HCO_3 levels decrease.

	pH	PCO_2	HCO_3^-
I	Normal	Increased	Decreased
II	Increased	Decreased	Increased
III	Decreased	Increased	Normal
IV	Increased	Increased	Increased
V	Decreased	Decreased	Normal

Question #: 5

A 46-year-old woman comes to the physician because of a 2-week history of yellowing of her skin and eyes. She also has pale and bulky feces. Physical examination shows scleral icterus. Laboratory studies show elevated serum alkaline phosphatase levels. Abdominal ultrasonography shows dilation of the biliary tree and the presence of gall stones in the common bile duct. Which of the following sets of laboratory results will most likely be found in this patient?

- A. I
- B. II
- C. III
- D. IV
- ✓E. V

Rationale: The patient has obstructive (cholestatic jaundice). Note the elevated serum ALP levels and presence of jaundice and steatorrhea. Dilation of the biliary tree confirms extrahepatic cholestasis. The direct (conjugated) bilirubin fraction would be elevated. Presence of gall stones prevents the entry of conjugated bilirubin into the intestine. Hence, the formation of stercobilinogen and urobilinogen is decreased. Absent urobilinogen in urine indicates complete obstruction.

Conjugated (water-soluble) bilirubin is found in urine and the urine has a deep orange-yellow color.

Lab findings III: Serum Unconjugated (indirect bilirubin) is elevated; Urine bilirubin absent and Urobilinogen elevated in urine: These are the lab findings in hemolytic (prehepatic) jaundice.

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Unconjugated	Absent	Absent
II	Unconjugated	Present	Increased
III	Unconjugated	Absent	Increased
IV	Conjugated	Absent	Increased
V	Conjugated	Present	Absent

Question #: 6

A 20-year-old man comes to the physician because of a 1-day history of passing brown-colored urine, muscle pain and muscle cramps after running on the treadmill for about 30 minutes. He says he can tolerate mild exercise. Examination of muscle biopsy specimen shows excessive glycogen accumulation. The structure and branching pattern of glycogen is normal. Muscle triglyceride levels are normal. Examination of a liver biopsy specimen shows normal glycogen structure and content. Which of the following is the most likely diagnosis?

- ✓A. McArdle disease
- B. Myopathic fatty acid oxidation defect

- C. Medium chain acyl CoA dehydrogenase deficiency
- D. vonGierke disease
- E. Duchenne muscular dystrophy

Rationale: The patient has a glycogen storage disorder affecting the muscle (symptoms and laboratory findings). An important test used in the diagnosis is the measurement of blood lactate levels following ischemic exercise. Blood lactate levels increase following ischemic exercise in normal persons due to glycogenolysis and anaerobic glycolysis. When there is a defect in glycogen breakdown (McArdle disease), there is NO increase in blood lactate levels following exercise.

In myopathic fatty acid oxidation defect, patients are unable to tolerate aerobic and sustained exercise. There is accumulation of lipid droplets in muscle.

vonGierke disease (liver glycogen storage disorder) and MCAD deficiency (systemic beta oxidation defect) are characterized by hypoglycemia.

Question #: 7

An investigator is studying the regulation of acetyl CoA carboxylase. It is noted that the presence of citrate activates the enzyme. Which of the following changes to the enzyme is most likely observed under these conditions?

- ✓A. Polymer formation
- B. Phosphorylation
- C. Dephosphorylation
- D. Monomer formation
- E. Up-regulation and increased synthesis

Rationale: Acetyl CoA carboxylase is the regulated enzyme of fatty acid synthesis. It converts acetyl CoA to malonyl CoA. Requires biotin as a coenzyme.

Following a carbohydrate-rich meal, glucose is converted to pyruvate which then forms acetyl CoA by the pyruvate dehydrogenase complex. Acetyl CoA is converted to citrate which is next transported to the cytosol for fatty acid synthesis. Presence of citrate facilitates polymerization and activation of the acetyl CoA carboxylase enzyme. Citrate is also an allosteric activator of the enzyme.

Palmitoyl CoA (end product of fatty acid synthesis) facilitates formation of the inactive monomer. It is a feedback inhibitor.

Acetyl CoA carboxylase is inhibited by phosphorylation by AMP-kinase. Glucagon activates AMP-kinase.

Dephosphorylation in the presence of insulin (fed state) activates the enzyme.

Insulin and high carbohydrate diets induce the gene for the enzyme resulting in increased synthesis of the enzyme (long-term regulation).

Question #: 8

A 9-year-old girl is brought to the physician by her parents because of a 2-week history of painful muscle cramps during exercise. Laboratory studies show normal plasma glucose level but no increase of lactate in the blood following exercise. Examination of a muscle biopsy specimen shows increased glycogen content with the normal pattern of branching in its structure. Inherited deficiency of which of the following is the most likely diagnosis?

- A. Carnitine palmitoyl transferase-II (CPT-II)
- B. Glycogen debranching enzyme
- C. Glycogen branching enzyme
- ✓D. Muscle glycogen phosphorylase
- E. Pyruvate dehydrogenase complex
- F. Medium chain acyl CoA dehydrogenase (MCAD)

Rationale: The vignette describes a patient with McArdle disease. Characterized by deficiency of the muscle isoenzyme of glycogen phosphorylase. The glycogen degradation in the muscle is impaired. There is reduced formation of glucose 1-phosphate and reduced entry of glucose 6-phosphate into glycolysis and reduced formation of lactate.

An important lab finding is NO increase in lactate following ischemic anaerobic activity.

Muscle biopsy shows increased glycogen accumulation and the structure of glycogen is normal.

In branching enzyme deficiency (Andersen disease), glycogen with long branches (Amylose-like) accumulates especially in the liver. Presents as cirrhosis in children.

In debranching enzyme deficiency (Cori disease), glycogen with short outer branches (Limit dextrin) accumulates in liver and muscle. Characterized by hepatomegaly and myopathy.

CPT-II deficiency presents as muscle cramps during aerobic activity. These patients are able to perform anaerobic exercise. Blood lactate levels increase (as in normal people) following ischemic exercise. This is important for differential diagnosis. Glycogen content in muscle biopsy is normal but there is accumulation of (TAG) lipid droplets in muscle.

MCAD deficiency is characterized by hypoglycemia and hypoketosis in infants. Metabolic crisis episodes occur following an infection or stress.

Question #: 9

A 50-year-old man comes to the physician because of a 3-week history of yellowing of the eyes. Physical examination shows jaundice and ascites. Results of laboratory studies are shown. Which of the following organs are also most likely affected in this disorder?

- A. Lungs
- B. Basal ganglia, cornea
- ✓C. Pancreas, skin
- D. Blood vessels and gums
- E. Twisted hair, blood vessel aneurysms
- F. Pancreas and lungs

Rationale: The vignette shows a patient with hemochromatosis. The liver is affected due to chronic iron accumulation. Elevated serum ferritin levels are a characteristic laboratory finding. Other organs affected are the skin (bronzing of skin), beta cells of pancreas (secondary diabetes), joints (joint pain) and heart (cardiomyopathy). These patients have unregulated and excessive dietary iron absorption due to impaired intestinal mucosal iron sensing mechanisms. An important differential diagnosis is Wilson disease (copper overload) which is characterized by liver damage, Kayser Fleischer ring in cornea and deposition of copper in the basal ganglia of brain. Serum ceruloplasmin levels are very low. Characterized by a defect in incorporation of copper to ceruloplasmin and impaired excretion of copper in bile. Urinary copper excretion is usually elevated in these patients.

Alpha 1- antitrypsin deficiency can affect the lungs (COPD) and liver (in some patients).

Blood vessels and gums are affected in vitamin C deficiency. But the liver is NOT affected.

Menke disease is characterized by poor absorption of dietary copper. Typically presents in infancy. Characterized by kinky hair, hypopigmentation and aneurysms. The liver is NOT affected in Menke disease.

Patients with cystic fibrosis have repeated respiratory infections and malabsorption due to block of exocrine pancreatic secretions by the viscid secretions. The liver is NOT commonly affected.

	Patient	Normal range
Alanine aminotransferase (ALT)	180	40-60
Aspartate aminotransferase (AST)	250	40-60
Alkaline phosphatase (ALP)	140	50-70
Gamma glutamyl transferase (GGT)	130	50-70
Serum ferritin	340	20-250 ng/mL

Question #: 10

A 32-year-old woman is brought to the emergency department by her roommate because of a 2-hour history of confusion and vomiting. She has a history of attempted suicide in the previous 6 months. Physical examination shows abdominal tenderness. Gastric lavage analysis shows an overdose of acetaminophen. Laboratory studies show abnormal liver function. Which of the following laboratory results is most likely observed in this patient?

- A. Elevated blood urea levels
- B. Elevated serum alpha fetoprotein levels
- ✓C. Elevated serum alanine aminotransferase levels
- D. Decreased serum alkaline phosphatase levels
- E. Serum AST: ALT ratio of 2:1

Rationale: The patient has drug induced acute hepatocellular damage. Highly elevated serum ALT levels are a characteristic finding in these patients.

Question #: 11

Six miners are admitted to the hospital after being trapped in a collapsed mine for one week with only water and no food. Laboratory studies show an average fasting plasma glucose of 70 mg/dL. Which of the following metabolic pathways is most likely responsible for the maintenance of the plasma glucose levels in these patients?

- A. Triacylglycerol hydrolysis in adipose tissue

- B. Glycogenolysis in the liver
- C. Ketone body synthesis in the liver
- ✓D. Gluconeogenesis in the liver
- E. Glycolysis in erythrocytes

Rationale: In prolonged fasting, gluconeogenesis from amino acids in the liver and kidneys is responsible for maintenance of blood glucose levels. Liver glycogen stores are completely depleted 24 hours following a fast.

Question #: 12

A 25-day-old boy is brought to the emergency department by his mother because of a 2-day history of drowsiness and progressively increasing unresponsiveness. Laboratory studies show respiratory alkalosis and a deficiency of carbamoyl phosphate synthetase-I (CPS-I). Which of the following additional laboratory findings is most likely in this patient?

- A. Elevated orotic acid in urine
- B. Elevated arginino succinic acid levels
- C. Low plasma glutamine levels
- ✓D. Elevated blood ammonia levels
- E. Elevated urinary C-8 and C10 acyl carnitine
- F. Elevated blood urea levels
- G. Elevated homocysteine levels

Rationale: The vignette describes a child with a urea cycle defect (hyperammonemia-I) due to deficiency of carbamoyl phosphate synthetase-I (CPS-I).

This is the most severe of the urea cycle defects. Blood ammonia levels and glutamine levels are highly elevated.

Blood urea levels are low. Low levels of all the urea cycle intermediates is a characteristic finding (citrulline, argininosuccinate).

Orotic acid levels are elevated in OTC deficiency (Ornithine transcarbamylase; Hyperammonemia type-II)

Urinary C-8 and C-10 acyl carnitine levels are elevated in systemic beta oxidation defect (MCAD deficiency).

Plasma homocysteine levels are elevated in homocystinuria due to deficiency of cystathionine synthase. Patients have arterial wall defects (ballooning and aneurysm of blood vessels), dislocation of the lens and skeletal deformities.

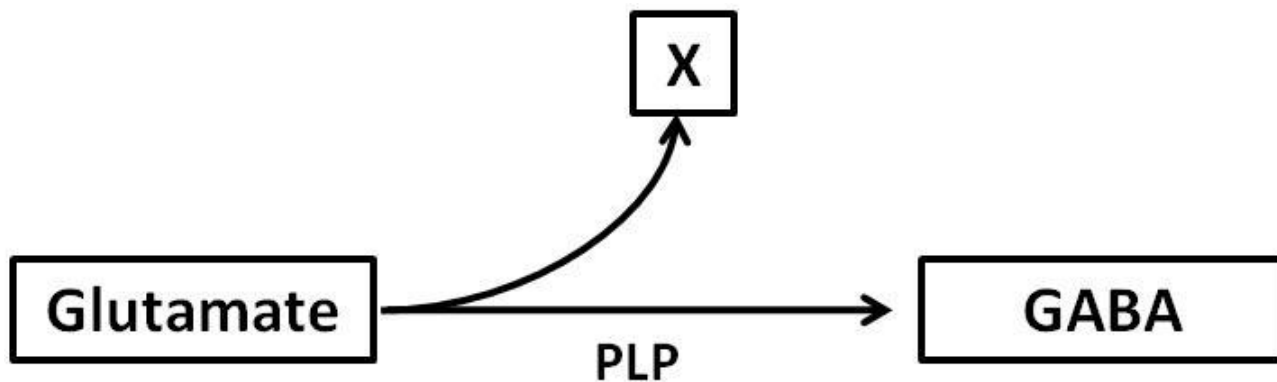
Homocysteine levels are also elevated in folic acid deficiency, pyridoxine deficiency and vitamin B12 deficiency.

Question #: 13

An investigator is studying the formation of a specific inhibitory neurotransmitter shown in the figure. Which of the following best describes the type of reaction shown in the figure?

- A. Carboxylation
- B. Transamination
- ✓C. Decarboxylation
- D. Dehydrogenation
- E. Oxidative deamination

Rationale: The conversion of glutamate to GABA is an example of an amino acid decarboxylation reaction. These reactions require PLP (vitamin B6) as a coenzyme. Review other examples of amino acid decarboxylation reactions - Dopamine formation, Serotonin formation, Histamine formation.



Question #: 14

A 10-day-old female newborn is brought to the physician by her mother for a follow-up examination because of a 1-week history of yellowing of the skin. Laboratory studies initially showed total bilirubin levels of 4mg/dL (Normal: 0.1-1 mg/dL) and direct bilirubin was not detected. Today, her laboratory studies show normal bilirubin levels. Which of the following is the best explanation for her symptoms?

- A. A temporary deficiency of heme oxygenase
- ✓B. A low activity of UDP-glucuronyl transferase
- C. Dubin-Johnson syndrome
- D. Crigler-Najjar syndrome type-I
- E. An abnormally low amount of serum albumin

Rationale: The vignette describes a child with physiological jaundice. Jaundice in the newborn is due to low activity of UDP-glucuronyl transferase and due to increased rates of hemolysis in the first few days of life.

Question #: 15

A 2-year-old girl is brought to the emergency department by her parents because of a sudden-onset of seizures. She has had similar incidents over the previous six months. Results of laboratory studies on admission are shown in the table. Which of the following inherited enzyme deficiency disorders is most likely in this patient?

- ✓A. Glucose 6-phosphatase
- B. Muscle glycogen phosphorylase
- C. Medium chain acyl CoA dehydrogenase deficiency
- D. Pyruvate dehydrogenase complex deficiency
- E. Aldolase B

Rationale: The child in the vignette most likely has von Gierke disease. The child has lactic acidosis and hypoglycemia. Children with PDH complex deficiency have lactic acidosis but blood glucose levels are usually within normal limits. Children with MCAD deficiency have hypoglycemia, low levels of ketone bodies but no lactic acidosis.

	Patient results	Reference ranges
pH	7.3	7.36-7.42
HCO ₃ ⁻	20 mEq/L	24-26 mEq/L
Fasting blood glucose	50 mg/dL	70-90 mg/dL
Serum lactate	Increased	-NA-

Question #: 16

An investigator studying muscle metabolic changes during extreme exertion and strenuous activity observes high levels of intracellular AMP. This results in the allosteric activation of various skeletal muscle enzymes. In which of the following skeletal muscle enzymes is this effect most likely observed?

- ✓A. Phosphofructokinase-1 and glycogen phosphorylase
- B. Hexokinase and glycogen phosphorylase
- C. Glycogen synthase and glycogen phosphorylase
- D. Debranching enzyme and pyruvate kinase
- E. Lactate dehydrogenase and phosphofructokinase-1
- F. Phosphofructokinase-1 and glycogen phosphorylase kinase

Rationale: AMP allosterically activates glycogen phosphorylase of glycogenolysis and PFK-1 of glycolysis. Glycogenolysis in muscle forms glucose 6-phosphate which then enters glycolysis to form lactate (when performing strenuous activity)

Question #: 17

A 35-year-old woman comes to the physician because of a 2-week history of easy bruising. Four months ago, she was diagnosed with gall stones 4 months ago. Stool examination shows steatorrhea. Laboratory studies show increased direct bilirubin levels and prolonged prothrombin time. Which of the following enzyme activities is most likely impaired in this patient?

- A. Methylmalonyl CoA mutase
- B. Propionyl CoA carboxylase
- ✓C. γ -glutamyl carboxylase in the liver
- D. 25-hydroxylase in the liver
- E. 1-hydroxylase in the kidney
- F. Prolid hydroxylase

Rationale: The vignette describes a patient with fat-soluble vitamin K deficiency (due to fat malabsorption). Vitamin K is a coenzyme for gamma-carboxylase in the liver. It is required for the gamma-carboxylation of clotting factors II, VII, IX and X.

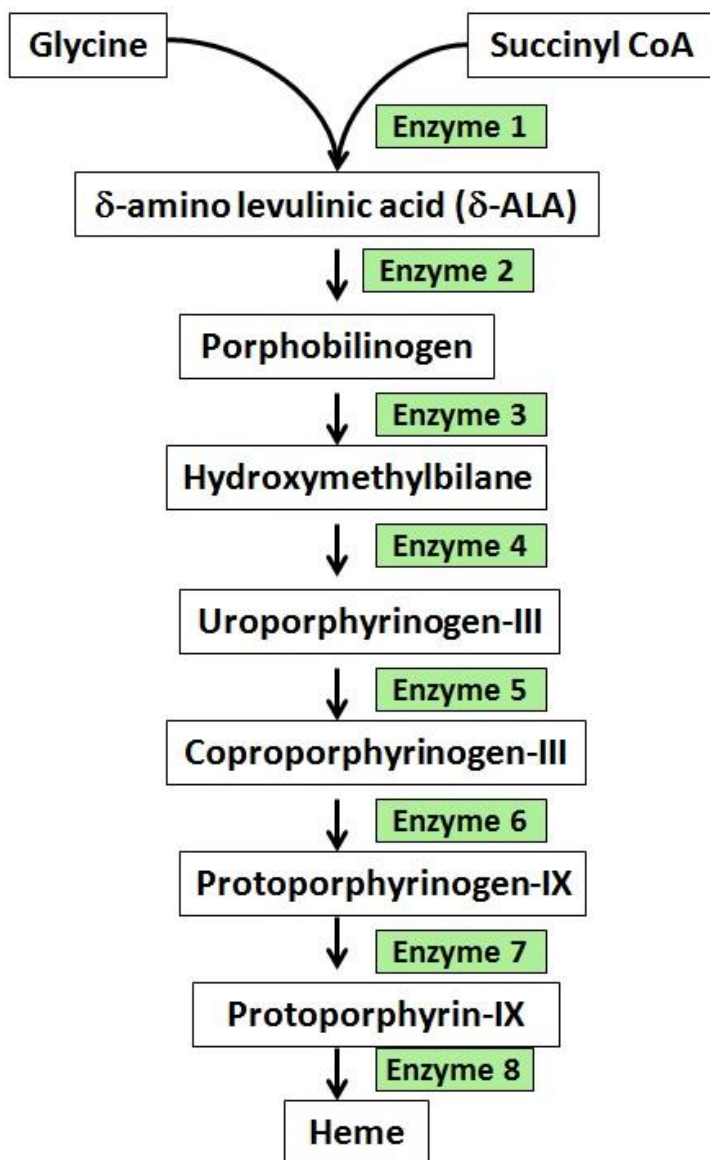
Question #: 18

An investigator is studying the pathway of heme synthesis shown in the figure in

the liver and bone marrow. Which of the following best describes this pathway?

- A. Enzyme 6 is inhibited by lead
- ✓B. Enzyme 1 requires PLP as a coenzyme
- C. Inherited deficiency of Enzyme 3 results in photosensitivity
- D. Enzyme 5 deficiency causes congenital erythropoietic porphyria
- E. Inherited deficiency of Enzyme 4 causes neuropsychiatric manifestations

Rationale: The pathway shown is the heme biosynthesis from glycine and succinyl CoA. Enzymes 2 and 8 are inhibited by lead. Deficiency of Enzyme 3 results in Acute intermittent porphyria (no photosensitivity). Enzyme 4 deficiency results in Congenital erythropoietic porphyria characterized by photosensitivity. Delta ALA synthase (enzyme 1) requires PLP as a coenzyme.



Question #: 19

A 24-year-old woman had a breakfast of cereal, coffee with sugar, toast, fried bacon and eggs. Liver metabolism is observed one hour later. Which of the following metabolic effects is most likely observed in the hepatocytes?

- A. Increased glycogen phosphorylase activity
- ✓B. Increased acetyl CoA carboxylase activity
- C. Decreased phosphofructokinase-1 activity
- D. Increased carnitine palmitoyl transferase-I activity
- E. Increased pyruvate carboxylase activity
- F. Increased mitochondrial HMG-CoA synthase activity

Rationale: The woman is in the well fed state (under the influence of insulin). Fatty acid biosynthesis is active. Liver glycogenolysis, gluconeogenesis, beta-oxidation and ketogenesis are less active. Liver glycolysis is active.

Question #: 20

An investigator is studying metabolism in experimental rats under different metabolic conditions. When fed a diet rich in carbohydrates, large amounts of citrate are formed in the cytosol of hepatocytes of experimental animals. Which of the following is the most likely effect of this intermediate on lipid metabolism?

- A. Activation of the carnitine shuttle system
- ✓B. Activation of fatty acid synthesis
- C. Inhibition of carnitine palmitoyl transferase-I (CPT-I)
- D. Inhibition of cholesterol biosynthesis
- E. Activation of bile acid synthesis
- F. Activation of ketogenesis

Rationale: The presence of high amounts of cytosolic citrate activates fatty acid biosynthesis by increasing the activity of acetyl CoA carboxylase. Presence of citrate results in the polymerization of the enzyme and activating it.

Question #: 21

A 24-year-old man fasts for 2 days because he wants to reduce his weight for an

upcoming modeling show. He takes only sips of orange juice equal to about 150 ml per day. Which of the following metabolic changes would most likely be present in the liver in this individual?

- A. Dephosphorylation of the PFK-2/FBPase-2 bifunctional enzyme
- ✓B. Induction of phosphoenolpyruvate carboxykinase
- C. Dephosphorylation of glycogen phosphorylase kinase
- D. Inactivation of protein kinase A
- E. Dephosphorylation of glycogen phosphorylase
- F. Increased levels of fructose 2,6 bisphosphate
- G. Increased activity of glycogen synthase

Rationale: The key enzymes of gluconeogenesis would be induced by glucagon in the fasting state. Review the other metabolic changes associated with the fasting state.

Question #: 22

A 36-year-old woman comes to the physician because of a 2-month history of intermittent abdominal pain. Physical examination shows tenderness over the right upper quadrant of the abdomen. Results of laboratory studies are shown. Which of the following is the most likely diagnosis?

- A. Hemolytic anemia
- B. Acute viral hepatitis
- C. Chronic alcoholic cirrhosis
- ✓D. Gall stones in the common bile duct
- E. Crigler Najjar syndrome type-I
- F. Gilbert syndrome

Rationale: The lab findings are characteristic of obstructive jaundice. Note the presence of conjugated hyperbilirubinemia, presence of bilirubin in urine. The absence of fecal stercobilin and urinary urobilinogen indicate complete obstruction of the biliary duct. Additional lab findings are elevated ALP and GGT levels.

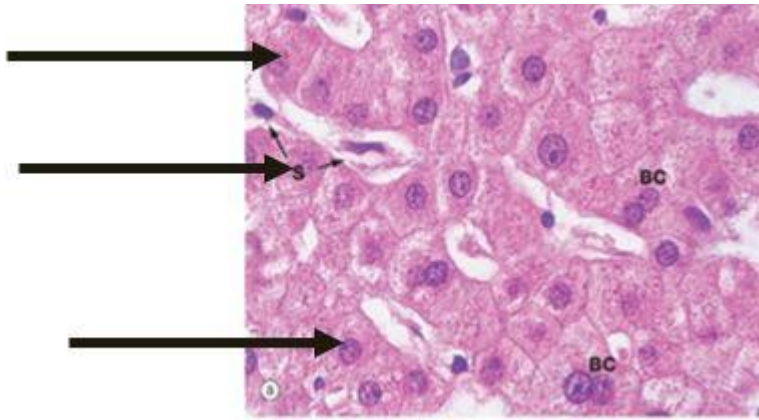
	Patient results	Reference results
Total serum bilirubin	6	1-2 mg/dL
Direct serum bilirubin	5.5	<0.2 mg/dL
Urine bilirubin	Present	Absent
Urine urobilinogen	Absent	Present
Fecal stercobilin	Absent	Present

Question #: 23

A 33-year-old woman comes to the physician because of a 1-week history of loss of appetite, nausea, weakness, and dark urine. Her temperature is 38°C (101°F), pulse is 100/min and respirations are 22/min. Physical examination shows right upper quadrant abdominal tenderness. Laboratory studies show antibodies against hepatitis B virus. A diagnosis of acute hepatitis B infection is made. In this condition, there is significant inflammation of the cells shown at the tips of the arrows in the attached photomicrograph. Which of the following is the most likely function of these cells?

- ✓A. Production of bilirubin
- B. Phagocytosis of red blood cells
- C. Storage of vitamin A as retinyl esters
- D. Concentration of bile
- E. Synthesis of secretin

Rationale: Hepatocytes carry out many functions, including production of bilirubin.



Question #: 24

A research group is investigating the role of bacteria in inflammatory bowel diseases of the gastrointestinal tract. They have identified a receptor that binds Type I pili on a subset of Gram-negative enterobacilli such as *Escherichia coli*. The group hypothesizes that the receptor is important for immunosurveillance at the mucosal surface and the initiation of an effective adaptive immune response. It appears that the receptor may be a specific marker for a particular cell type. Which of the following best characterizes the cell type most likely involved in this function?

- A. Acidophilic apical secretory vesicles
- B. Columnar cells with a striated border
- C. Produces hormones that act on the gallbladder and pancreas
- D. Large accumulations of mucinogen granules in the apical cytoplasm
- ✓E. Abundant microfolds on the apical surface

Rationale: E - The M-cells are found at the tips Peyer's patches, facing the lumen of the intestine. They act as antigen transporting cells, where they take antigens from the lumen and present it to the immune cells in Peyer patches.

- A - paneth cells - they also have a basophilic basal cytoplasm
- B - enterocytes, which are the most abundant epithelial cells. They have microvilli on their apical surface because they are involved in absorption
- C- enteroendocrine cells
- D - goblet cells

Question #: 25

A 56-year-old man comes to the physician because of a 2-week history of heartburn. Physical examination shows epigastric tenderness. Laboratory studies show no abnormalities. An endoscopic exam is requested, and multiple biopsies are taken during this procedure. One of the light micrographs of the biopsy specimen is attached for your reference. Which of the following histological features helps to identify the segment where the biopsy was taken from?

- A. Intestinal crypts of Lieberkühn
- B. Esophageal glands
- C. Von Ebner's glands
- D. Enteroendocrine glands
- ✓E. Brunner's glands

Rationale: Glands in the submucosa are only found in the esophagus and the duodenum of the gastrointestinal tract. In all other areas, the glands are limited to the lamina propria. The photomicrograph shows villi, which is a characteristic feature of the small intestine. Therefore villi with submucosal glands is the Duodenum, where the submucosal glands are specifically called Brunner's glands



Question #: 26

A 52-year-old woman comes to the physician because of a 3-day history of diarrhea. She has been drinking only small amounts because she feels nauseous. Physical examination shows dry lips and increased capillary refill times. She is found to have hypovolemia and needed fluid replacement. She receives an oral rehydration solution containing NaCl, glucose, KCl and citrate dissolved in water. Which of the following intestinal uptake mechanisms is most likely targeted by this rehydration solution to restore fluid in the patient?

- A. Sodium-chloride co-transport
- B. Passive sodium absorption
- C. Chloride absorption via CFTR channels
- ✓D. Sodium-glucose co-transport
- E. Active potassium absorption

Rationale: The intestinal SGLT-1 uses sodium to co-transport glucose, along with water. This is one of the most high-volume intestinal uptake systems and therefore targeted by ORS, which is why they include glucose.

Question #: 27

A 3-week-old boy is brought to the physician by his mother for a routine examination. The mother says that his bowel movements increase after every feeding. Physical examination shows no abnormalities. The physician reassures the mother not to worry as this is physiological. Which of the following reflexes best explains the cause of this physiological occurrence in the newborn?

- A. Gastroileal
- ✓B. Gastrocolic
- C. Intestino-intestinal
- D. Defecation
- E. Peristaltic rush

Rationale: The gastrocolic reflex is a reflex in which the simple act of eating stimulates movement in the gastrointestinal tract. Specifically, the gastrocolic reflex stimulates contractions in the large intestine. These contractions may lead to a bowel movement a short time after eating, which is consistent with the description in the vignette. Accordingly, the gastrocolic reflex

is the correct answer. A quick look at the other responses confirms this initial intuition: gastroileal is incorrect since it affects mainly the IC valve motility in the small intestine, which is not directly related to bowel movements. Intestino-intestinal is also incorrect because it refers to a short reflex where distension of one part of the intestine leads to decreased contractions just caudal of the chyme bolus, again not directly relevant to inducing large bowel movements. Defecation is not a reflex, just another word for bowel movements. Peristaltic rush refers to a rapid movement of intestinal contents that results in diarrhea, caused by an absence of peristaltic rhythm and sphincter tone, possibly triggered by severe GI irritation, which is also not consistent with the reference to the 'normal occurrence' of these bowel movements in the vignette.

Question #: 28

A 3-year-old girl is brought to the physician by her mother because of a 2-week history of diarrhea characterized by pale, bulky and foul-smelling stools. Her mother is also concerned about her failure to thrive. Laboratory studies show elevated sweat chloride concentrations. Genetic studies show CFTR gene mutation. Decreased pancreatic secretion of which of the following best explains diarrhea in this patient?

- ✓A. Lipase
- B. Secretin
- C. Amylase
- D. Cholecystokinin
- E. Bicarbonate

Rationale: Hereditary conditions such as cystic fibrosis can cause pancreatitis. The pancreas becomes scarred and loses its ability to make enough digestive enzymes. Lack of enzymes due to pancreatic damage results in poor digestion and absorption of food, especially fats. In the digestive tract, TAG is hydrolyzed by the enzyme pancreatic lipase, to release free fatty acids and monoglycerides. Patients may notice bulky smelly bowel movements due to too much fat (steatorrhea). Occasionally, an "oil slick" can be seen on the toilet water.

Question #: 29

A 28-year-old man comes to the clinic because of chronic diarrhea. He says his daily stool output is approximately 2 liters (tenfold increase). He has just recently undergone a surgical resection of his terminal ileum because of Crohn's disease. Which of the following substances are most likely responsible for his current

condition?

- ✓A. Bile acids
- B. Inflammatory cytokines
- C. Short chain fatty acids
- D. Prostaglandins
- E. Vasoactive intestinal polypeptides

Rationale: (A) Bile acid diarrhea (BAD), also known as bile acid malabsorption (BAM), is a condition in which bile acids are either not properly re-absorbed in small intestine, or too much bile is produced. Significant amounts of bile acids then reach the large intestine, which leads to increased fluid secretion, resulting in watery stools and increased intestinal motility, both of which lead to symptoms of diarrhea. The condition is known to co-occur with IBS and Crohn's disease, especially after ileocecal resection. None of the other answers are very plausible direct causes of diarrhea, so (A) is the best answer.

Question #: 30

A 45-year-old woman comes to the physician because of a 2-month history of fatigue, weight loss, bone pains and diarrhea. She says she passes a stool that is sticky and foul-smelling. She has a 10-year history of Crohn's disease and underwent an ileal resection 3 months ago. Her medical records show that 60 cm of distal ileum was removed during the surgery. Absorption of which of the following substances most likely are affected following the surgery in this patient?

- A. Vitamin B12 and calcium
- ✓B. Vitamin B12 and bile acids
- C. Calcium and fat-soluble vitamins
- D. Fat-soluble vitamins and glucose
- E. Bile acids and sodium

Rationale: Distal ileal resection affects absorption of mainly 2 substances, bile acids and vit B12 because these 2 substances are absorbed only in this region of the gut. Fat absorption and fat-soluble vitamin absorption are also affected to some extent, but not critically affected as only 60-cm length of the intestine is resected.